

指针与抽象

结构体、指针与STL容器 · 10题 · C++ & Python 双语对照

NQ091 AcWing 16 AcWing 16

AcWing 16

- 输入** 见原题
- 输出** 见原题
- 范围** 见原题

输入	输出
见原题	见原题

思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
class Solution {
public:
    string replaceSpaces(string &str) {
        string res;
        for (auto c : str)
            if (c == ' ') res += "%20";
            else res += c;
        return res;
    }
};
```

Python

```
# AcWing 16
```

NQ092 AcWing 17 AcWing 17

AcWing 17

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思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     ListNode *next;
 *     ListNode(int x) : val(x), next(NULL) {}
 * };
 */
class Solution {
public:
    vector<int> printListReversingly(ListNode*
head) {
        vector<int> res;
        if (!head) return res;
        auto p = head;
        while(p) {
            res.push_back(p->val);
            p=p->next;
        }
        reverse(res.begin(), res.end());
        return res;
    }
};
```

Python

AcWing 17

NQ093 AcWing 20 AcWing 20

AcWing 20

输入 见原题

输出 见原题

范围 见原题

输入

见原题

输出

见原题

思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```

class MyQueue {
public:
    stack<int> stk, cache;
    /** Initialize your data structure here. */
    MyQueue() {

    }

    /** Push element x to the back of queue. */
    void push(int x) {
        stk.push(x);
    }

    void copy(stack<int>& a, stack<int>& b){
        while (a.size()) {
            b.push(a.top());
            a.pop();
        }
    }

    /** Removes the element from in front of queue and returns that element. */
    int pop() {
        copy(stk,cache); //拷贝到cache数组
        int res = cache.top();
        cache.pop();
        copy(cache,stk);
        return res;
    }

    /** Get the front element. */
    int peek() {
        copy(stk,cache); //拷贝到cache数组
        int res = cache.top();
        copy(cache,stk);
        return res;
    }

    /** Returns whether the queue is empty. */
    bool empty() {
        return stk.empty();
    }
};

/**
 * Your MyQueue object will be instantiated and called as such:
 * MyQueue obj = MyQueue();
 * obj.push(x);
 * int param_2 = obj.pop();
 * int param_3 = obj.peek();
 * bool param_4 = obj.empty();
 */

```

Python

AcWing 20

NQ094 AcWing 21 AcWing 21

AcWing 21

输入

见原题

输出

见原题

范围

见原题

输入
见原题

输出
见原题

思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
class Solution {
public:
    int Fibonacci(int n) {
        if (n <= 1) return n;
        return Fibonacci(n - 1) + Fibonacci(n - 2);
    }
};
```

Python

```
# AcWing 21
```

NQ095 AcWing 35 AcWing 35

AcWing 35

输入

见原题

输出

见原题

范围

见原题

输入
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输出
见原题

思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```

/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     ListNode *next;
 *     ListNode(int x) : val(x), next(NULL) {}
 * };
 */
class Solution {
public:
    ListNode* reverseList(ListNode* head) {
        if (!head || !head->next) return head;

        auto p = head, q = p->next;
        while (q)
        {
            auto o = q->next;
            q->next = p;
            p = q, q = o;
        }
        head->next = NULL;

        return p;
    }
};

```

Python

AcWing 35

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AcWing 36

输入

 见原题

输出

 见原题

范围

 见原题

输入

见原题

输出

见原题

思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
/**
 * Definition for singly-linked list.
 * struct ListNode {
 *     int val;
 *     ListNode *next;
 *     ListNode(int x) : val(x), next(NULL) {}
 * };
 */
class Solution {
public:
    ListNode* merge(ListNode* l1, ListNode* l2)
    {
        auto dummy = new ListNode(-1), tail = dummy;
        while (l1 && l2)
            if (l1->val < l2->val)
            {
                tail->next = l1;
                l1 = l1->next;
            }
            else
            {
                tail->next = l2;
                l2 = l2->next;
            }

            if (l1) tail->next = l1;
            if (l2) tail->next = l2;

            return dummy->next;
    }
};
```

Python

AcWing 36

NQ097 AcWing 862 AcWing 862

AcWing 862

输入 见原题**输出** 见原题**范围** 见原题

输入
见原题

输出
见原题

思路 见AcWing原题描述**技巧** 参考模板库 CONTEXT.md

C++

```

#include <iostream>
#include <algorithm>

using namespace std;

const int N = 10010;

struct data {
    int x;
    double y;
    string z;
    bool operator< (const data & t) const
    {
        return x < t.x;
    }
} a[N];

int main() {
    int n;
    cin >> n;
    for (int i = 0; i < n; i++) cin >> a[i].x >> a[i].y >> a[i].z;
    // 结构体排序
    sort(a, a + n);
    // 输出结果
    for (int i = 0; i < n; i++)
        printf("%d %.2lf %s\n", a[i].x, a[i].y, a[i].z.c_str());
    return 0;
}

```

Python

AcWing 862

NQ098 AcWing 810

AcWing 810

AcWing 810

输入

 见原题

输出

 见原题

范围

 见原题

输入

见原题

输出

见原题

思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
#include <iostream>

using namespace std;

int abs(int x)
{
    if (x > 0) return x;
    return -x;
}

int main()
{
    int x;
    cin >> x;
    cout << abs(x) << endl;

    return 0;
}
```

Python

AcWing 810

NQ099 AcWing 814

AcWing 814

AcWing 814

输入

见原题

输出

见原题

范围

见原题

输入

见原题

输出

见原题

思路 见AcWing原题描述**技巧** 参考模板库 CONTEXT.md

C++

```

#include <iostream>
#include <cstring>

using namespace std;

const int N = 110;

void copy(int a[], int b[], int size)
{
    memcpy(b, a, size * 4);
}

int main()
{
    int a[N], b[N];
    int n, m, size;
    cin >> n >> m >> size;
    for (int i = 0; i < n; i++) cin >> a[i];
    for (int i = 0; i < m; i++) cin >> b[i];

    copy(a, b, size);

    for (int i = 0; i < m; i++) cout << b[i]
<< ' ';
    cout << endl;

    return 0;
}

```

Python

AcWing 814

NQ100 AcWing 816 AcWing 816

AcWing 816

 见原题

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输入

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输出

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思路 见AcWing原题描述

技巧 参考模板库 CONTEXT.md

C++

```
#include <iostream>

using namespace std;

void reverse(int a[], int size)
{
    for (int i = 0, j = size - 1; i < j; i ++, j -- )
        swap(a[i], a[j]);
}

int main()
{
    int a[1000];
    int n, size;

    cin >> n >> size;
    for (int i = 0; i < n; i ++ ) cin >> a[i];
    reverse(a, size);

    for (int i = 0; i < n; i ++ ) cout << a[i]
    << ' ';

    return 0;
}
```

Python

AcWing 816